

TrES-1b

R-0202

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_240815 · created 2026-07-16T22:35:44+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460537.7514 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.01
Transit depth (Rp/Rs)^2	2.06 +/- 0.29 [%]
Orbital Inclination (inc)	87.37 +/- 0.92
Airmass coefficient 1 (a1)	1.113 +/- 0.01
Airmass coefficient 2 (a2)	-0.106 +/- 0.06
Scatter in the residuals of the lightcurve fit is	0.93 %
Best Comparison Star	#3 - [612, 177]
Optimal Aperture	4.58
Optimal Annulus	13.95
Transit Duration (day)	0.0887 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.144 ± 0.01 vs 0.1358 (deviation 0.82σ)
Duration fit vs published	0.0887 d vs 0.1042 d (deviation 2.81σ)
Mid-time O–C	-7.3 min
Depth SNR	14.4
Residual scatter	0.93 %

Light curve

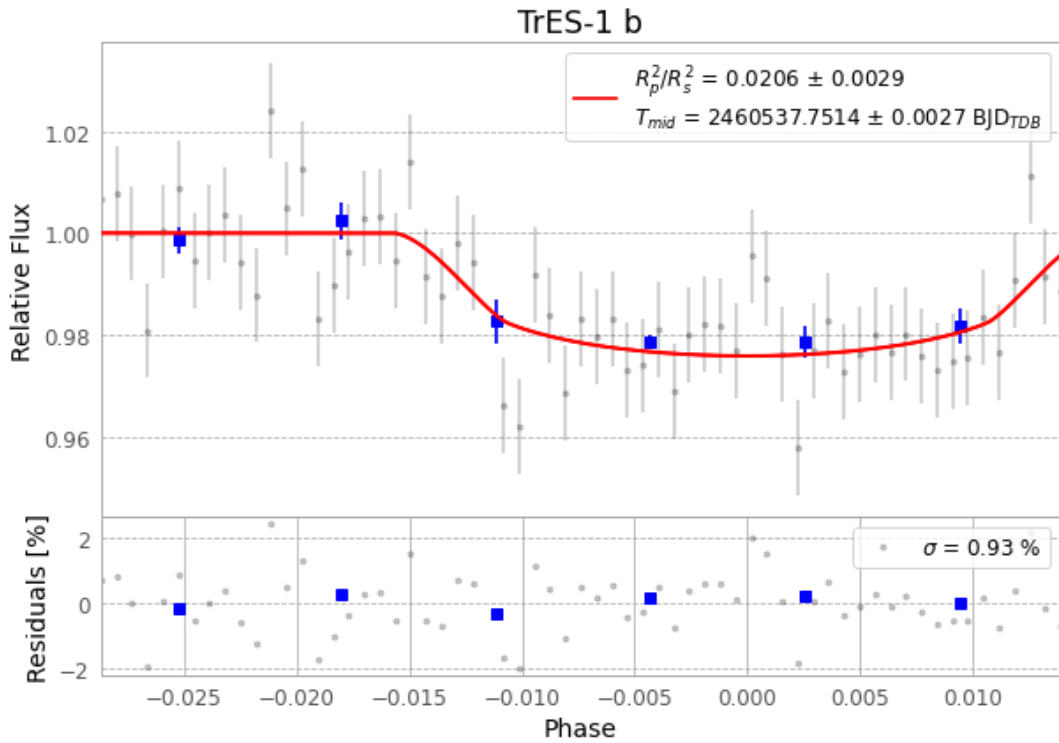


Figure 1 — FinalLightCurve_TrES-1 b_2024-08-14.png (SHA-256 f25694ee52d611ea...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [319, 198], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f4dfbf9748746023...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e

Data provenance

66 FITS frames (43 MB), TRES-1240815035115.FITS → TRES-1240815070615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-240815.log (cf431a13a7a2...), FinalLightCurve_TrES-1 b_2024-08-14.png (f25694ee52d6...), FinalLightCurve_TrES-1 b_2024-08-14.csv (6b72400ff101...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 22:35Z.